



# 1843 - Cloud composition and origin of the reddest known sub-stellar companion

## HD 206893 B

Cycle: 1, Proposal Category: GO

### INVESTIGATORS

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### OBSERVATIONS

| <i>Folder</i>                       | <i>Observation</i> | <i>Label</i> | <i>Observing Template</i>              | <i>Science Target</i> |
|-------------------------------------|--------------------|--------------|----------------------------------------|-----------------------|
| NIRISS/AMI of HD 206893 + companion |                    |              |                                        |                       |
|                                     | 1                  | SCI          | NIRISS Aperture Masking Interferometry | (1) HD-206893         |
|                                     | 2                  | CAL          | NIRISS Aperture Masking Interferometry | (2) HD-205827         |

### ABSTRACT

JWST will open up a new parameter space for studying exoplanet atmospheres in the near- and mid-infrared beyond  $\sim 3$  microns. The reddest known sub-stellar companion HD 206893 B represents an important and extreme test case for atmospheric models of exoplanets and brown dwarfs, and its unusual color sets constraints on the cloud composition and dust parameters which such models need to predict. Currently, no atmospheric model can

predict the extremely red color of HD 206893 B. We aim to study the composition and size distribution of the dust grains causing the red color of HD 206893 B. Moreover, we aim to constrain the C/O abundance ratio of HD 206893 B, which will in turn unravel its formation history and resolve the current uncertainty about its age and mass. Located at an angular separation of  $\sim 200$  mas from its host star and with a contrast of  $\sim 7.5$ - $8.5$  mag, we identify aperture masking interferometry with NIRISS as the only way to study this companion at  $\sim 4$ - $5$  microns with sufficient precision to discriminate dust grain properties and constrain its C/O abundance. L- and M-band observations with JWST are key for understanding the nature of HD 206893 B's atmosphere and the formation history of this enigmatic object. Ultimately, these observations will provide a benchmark test for improving atmospheric models of very cloudy and dusty sub-stellar objects and increase the reliability of such models in the  $\sim 4$ - $5$  microns regime which is hardly accessible from the ground.

## **OBSERVING DESCRIPTION**

This proposal aims to study the atmosphere of the reddest known sub-stellar companion HD 206893 B through precise photometric measurements at  $\sim 4$ - $5$  microns with NIRISS/AMI.

We will acquire data of HD 206893 and a PSF reference (HD 205827) in a SCI-CAL observing sequence. Both targets will be observed with the F480M, the F380M, and the F430M filter, in exactly this sequence in order to minimize the motion of the filter wheel between subsequent observations. During TA, we aim for a SNR  $> 100$  in order to achieve a centering accuracy of  $< 0.05$  pixels. The exposure times were computed based on the equation in JDox which states that a companion of contrast  $c$  can be detected with  $100/c^2$  photons, where  $0 < c < 1$  (cf. also Ireland 2013). The total time including overheads amounts to 2.47 hours (5167 sec for HD 206893 and 3714 sec for the PSF reference).

As recommended by JDox, we do not perform dithering. Dithering could lead to persistence on the detector and is not necessary for aperture masking interferometry.

# Proposal 1843 - Targets - Cloud composition and origin of the reddest known sub-stellar companion HD 206893 B

| Fixed Targets | #                                                                                                                                                                            | Name      | Target Coordinates                                                                   | Targ. Coord. Corrections                                                                                                                 | Miscellaneous |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|--------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|---------------|
|               | (1)                                                                                                                                                                          | HD-206893 | RA: 21 45 22.0050 (326.3416875d)<br>Dec: -12 47 0.07 (-12.78335d)<br>Equinox: J2000  | Proper Motion RA: 0.006410629200709497 sec of time/yr<br>Proper Motion Dec: 1.6E-5 arcsec/yr<br>Epoch of Position: 2015.5                |               |
|               | Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.<br>Category=Star<br>Description=[Brown dwarfs, Exoplanet Systems, F stars] |           |                                                                                      |                                                                                                                                          |               |
| Fixed Targets | (2)                                                                                                                                                                          | HD-205827 | RA: 21 38 16.5245 (324.5688521d)<br>Dec: -12 02 42.14 (-12.04504d)<br>Equinox: J2000 | Proper Motion RA: -9.79089087542528E-4 sec of time/yr<br>Proper Motion Dec: -0.025793999998313666 arcsec/yr<br>Epoch of Position: 2015.5 |               |
|               | Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.<br>Category=Calibration<br>Description=[Point spread function]             |           |                                                                                      |                                                                                                                                          |               |
|               |                                                                                                                                                                              |           |                                                                                      |                                                                                                                                          |               |

# Proposal 1843 - Observation 1 - Cloud composition and origin of the reddest known sub-stellar companion HD 206893 B

|                   |                                                                                                   |                 |                                                                                     |            |                    |                                                                                                                           |                    |                     |                     |                  |
|-------------------|---------------------------------------------------------------------------------------------------|-----------------|-------------------------------------------------------------------------------------|------------|--------------------|---------------------------------------------------------------------------------------------------------------------------|--------------------|---------------------|---------------------|------------------|
| Observation       | Proposal 1843, Observation 1: SCI                                                                 |                 |                                                                                     |            |                    |                                                                                                                           |                    |                     |                     |                  |
|                   | Diagnostic Status: Warning                                                                        |                 |                                                                                     |            |                    |                                                                                                                           |                    |                     |                     |                  |
| Diagnostics       | Observing Template: NIRISS Aperture Masking Interferometry                                        |                 |                                                                                     |            |                    |                                                                                                                           |                    |                     |                     |                  |
|                   | (Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.       |                 |                                                                                     |            |                    |                                                                                                                           |                    |                     |                     |                  |
| Fixed Targets     | #                                                                                                 | Name            | Target Coordinates                                                                  |            |                    | Targ. Coord. Corrections                                                                                                  |                    |                     | Miscellaneous       |                  |
|                   | (1)                                                                                               | HD-206893       | RA: 21 45 22.0050 (326.3416875d)<br>Dec: -12 47 0.07 (-12.78335d)<br>Equinox: J2000 |            |                    | Proper Motion RA: 0.006410629200709497 sec of time/yr<br>Proper Motion Dec: 1.6E-5 arcsec/yr<br>Epoch of Position: 2015.5 |                    |                     |                     |                  |
| Acquisition       | Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. |                 |                                                                                     |            |                    |                                                                                                                           |                    |                     |                     |                  |
|                   | Category=Star<br>Description=[Brown dwarfs, Exoplanet Systems, F stars]                           |                 |                                                                                     |            |                    |                                                                                                                           |                    |                     |                     |                  |
| Template          | #                                                                                                 | Target          | Acquisition Mode                                                                    | Filter     | Readout Pattern    | Groups/Int                                                                                                                | Integrations/Exp   | Total Integrations  | Total Exposure Time | ETC Wkbk.Calc ID |
|                   | 1                                                                                                 | SAME            | AMIBRIGHT                                                                           | F480M      | NISRAPID           | 5                                                                                                                         | 1                  | 1                   | 0.293               | 53691.1          |
| Dithers           | Subarray                                                                                          |                 |                                                                                     |            |                    | Direct Image                                                                                                              |                    |                     |                     |                  |
|                   | SUB80                                                                                             |                 |                                                                                     |            |                    | false                                                                                                                     |                    |                     |                     |                  |
| Spectral Elements | #                                                                                                 | Primary Dithers |                                                                                     |            | Subpixel Positions |                                                                                                                           |                    |                     |                     |                  |
|                   | 1                                                                                                 | NONE            |                                                                                     |            | NONE               |                                                                                                                           |                    |                     |                     |                  |
| PSF References    | #                                                                                                 | Filter          | Readout Pattern                                                                     | Groups/Int | Integrations/Exp   | Total Dithers                                                                                                             | Total Integrations | Total Exposure Time | ETC Wkbk.Calc ID    |                  |
|                   | 1                                                                                                 | F480M           | NISRAPID                                                                            | 11         | 720                | 1                                                                                                                         | 720                | 666.547             | 53691.2             |                  |
|                   | 2                                                                                                 | F380M           | NISRAPID                                                                            | 4          | 7046               | 1                                                                                                                         | 7046               | 2802.053            | 53691.3             |                  |
|                   | 3                                                                                                 | F430M           | NISRAPID                                                                            | 8          | 2177               | 1                                                                                                                         | 2177               | 1522.681            | 53691.4             |                  |
| PSF References    | CAL (Obs 2) (PSF Reference; Filters [F380M, F430M, F480M])                                        |                 |                                                                                     |            |                    |                                                                                                                           |                    |                     |                     |                  |
|                   | Additional Justification: false                                                                   |                 |                                                                                     |            |                    |                                                                                                                           |                    |                     |                     |                  |

Proposal 1843 - Observation 1 - Cloud composition and origin of the reddest known sub-stellar companion HD 206893 B

|                      |                                                                                                                                                           |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Special Requirements | <div data-bbox="147 105 457 126">Before Date 01-JAN-2023:00:00:00</div> <div data-bbox="147 147 518 168">Group Observations 1, 2, Non-interruptible</div> |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|

# Proposal 1843 - Observation 2 - Cloud composition and origin of the reddest known sub-stellar companion HD 206893 B

|                   |                                                                                                   |                 |                                                                                      |            |                  |                                                                                                                                          |                    |                     |                     |                  |
|-------------------|---------------------------------------------------------------------------------------------------|-----------------|--------------------------------------------------------------------------------------|------------|------------------|------------------------------------------------------------------------------------------------------------------------------------------|--------------------|---------------------|---------------------|------------------|
| Observation       | Proposal 1843, Observation 2: CAL                                                                 |                 |                                                                                      |            |                  |                                                                                                                                          |                    |                     |                     |                  |
|                   | Diagnostic Status: Warning                                                                        |                 |                                                                                      |            |                  |                                                                                                                                          |                    |                     |                     |                  |
|                   | Observing Template: NIRISS Aperture Masking Interferometry                                        |                 |                                                                                      |            |                  |                                                                                                                                          |                    |                     |                     |                  |
| Diagnostics       | (Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.       |                 |                                                                                      |            |                  |                                                                                                                                          |                    |                     |                     |                  |
| Fixed Targets     | #                                                                                                 | Name            | Target Coordinates                                                                   |            |                  | Targ. Coord. Corrections                                                                                                                 |                    |                     | Miscellaneous       |                  |
|                   | (2)                                                                                               | HD-205827       | RA: 21 38 16.5245 (324.5688521d)<br>Dec: -12 02 42.14 (-12.04504d)<br>Equinox: J2000 |            |                  | Proper Motion RA: -9.79089087542528E-4 sec of time/yr<br>Proper Motion Dec: -0.025793999998313666 arcsec/yr<br>Epoch of Position: 2015.5 |                    |                     |                     |                  |
|                   | Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. |                 |                                                                                      |            |                  |                                                                                                                                          |                    |                     |                     |                  |
|                   | Category=Calibration                                                                              |                 |                                                                                      |            |                  |                                                                                                                                          |                    |                     |                     |                  |
|                   | Description=[Point spread function]                                                               |                 |                                                                                      |            |                  |                                                                                                                                          |                    |                     |                     |                  |
| Acquisition       | #                                                                                                 | Target          | Acquisition Mode                                                                     | Filter     | Readout Pattern  | Groups/Int                                                                                                                               | Integrations/Exp   | Total Integrations  | Total Exposure Time | ETC Wkbk.Calc ID |
|                   | 1                                                                                                 | SAME            | AMIBRIGHT                                                                            | F480M      | NISRAPID         | 5                                                                                                                                        | 1                  | 1                   | 0.293               | 53691.6          |
| Template          | Subarray                                                                                          |                 |                                                                                      |            |                  | Direct Image                                                                                                                             |                    |                     |                     |                  |
|                   | SUB80                                                                                             |                 |                                                                                      |            |                  | false                                                                                                                                    |                    |                     |                     |                  |
| Dithers           | #                                                                                                 | Primary Dithers |                                                                                      |            |                  |                                                                                                                                          | Subpixel Positions |                     |                     |                  |
|                   | 1                                                                                                 | NONE            |                                                                                      |            |                  |                                                                                                                                          | NONE               |                     |                     |                  |
| Spectral Elements | #                                                                                                 | Filter          | Readout Pattern                                                                      | Groups/Int | Integrations/Exp | Total Dithers                                                                                                                            | Total Integrations | Total Exposure Time | ETC Wkbk.Calc ID    |                  |
|                   | 1                                                                                                 | F480M           | NISRAPID                                                                             | 10         | 641              | 1                                                                                                                                        | 641                | 545.055             | 53691.7             |                  |
|                   | 2                                                                                                 | F380M           | NISRAPID                                                                             | 3          | 7800             | 1                                                                                                                                        | 7800               | 2513.472            | 53691.8             |                  |
|                   | 3                                                                                                 | F430M           | NISRAPID                                                                             | 7          | 1885             | 1                                                                                                                                        | 1885               | 1176.24             | 53691.9             |                  |
| PSF References    | PSF Reference: true                                                                               |                 |                                                                                      |            |                  |                                                                                                                                          |                    |                     |                     |                  |

Proposal 1843 - Observation 2 - Cloud composition and origin of the reddest known sub-stellar companion HD 206893 B

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|----------------------|--------------------------------------------|
| Special Requirements | Before Date 01-JAN-2023:00:00:00           |
|                      | Group Observations 1, 2, Non-interruptible |